THE UNIVERSITY OF HONG KONG

BACHELOR OF ENGINEERING: LEVEL 1 EXAMINATION

DEPARTMENT OF CIVIL ENGINEERING

ENGINEERING MECHANICS AND MATERIALS – CIVL1013

DATE: 11 May, 2010 TIME: 9:30 am – 12:30 pm (3 hours)

Answer any 2 questions from Section A and any 2 questions from Section B.
All questions carry equal marks.
Use a separate answer book for Section A and Section B.

Use of Calculators:

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

SECTION A

You are required to answer 2 out of 3 questions in Section A.

Question 1

Consider the cantilever beam of constant flexural rigidity *EI*, shown in Figure 1, which is subjected to a linearly varying uniformly distributed load of maximum intensity $2w_o$ and minimum intensity w_o.

(a) Calculate all reactions. (**5 marks**)
(b) Derive the equation of the deflection curve with respect to distance x from the fixed support A. (**14 marks**)
(c) Calculate the deflection at support A and free end B. (**3 marks**)
(d) Calculate the rotation at support A and free end B. (**3 marks**)

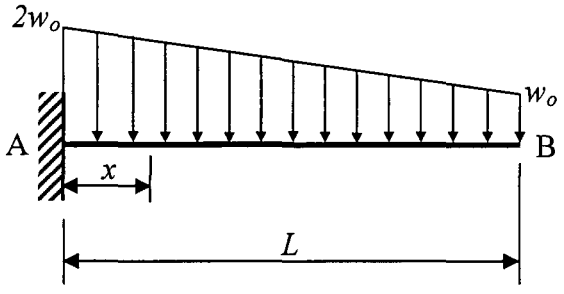


Figure 1

Question 2

Consider the rectangular simply supported beam with overhangs as shown in Figure 2.

(a) Calculate all reactions (include suitable check). (**6 marks**)
(b) Calculate the internal stress resultants of *V* and *M* at position B. (**4 marks**)
(c) Draw the shear force diagram for the entire beam. (**7 marks**)
(d) Draw the tension side bending moment diagram for the entire beam. (**8 marks**)

{note: show the position and magnitude of the maximum bending moment on the appropriate diagrams(s)}

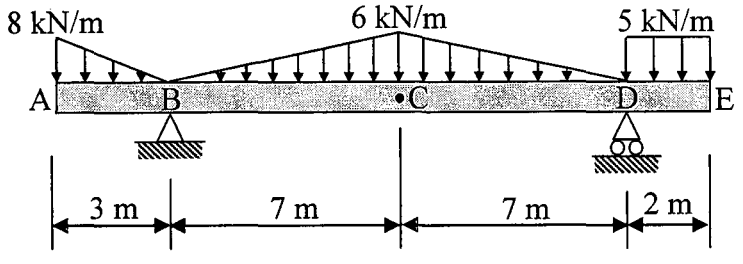


Figure 2

Question 3

(a) Consider the plane stress element given in Figure 3. Construct the Mohr's circle and then calculate the principal stresses (you do not need to calculate the orientation of the principal planes). (**9 marks**)

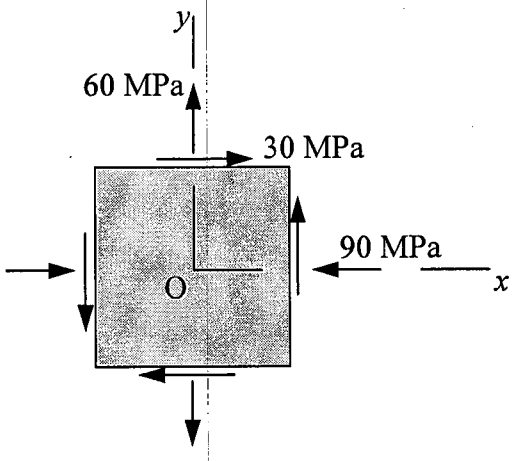
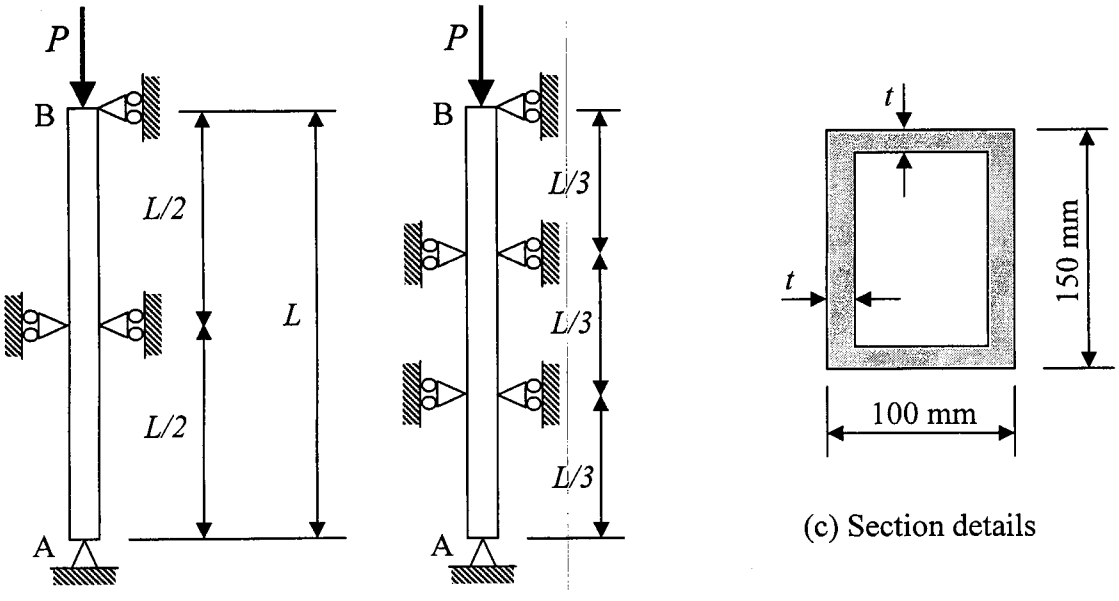


Figure 3

(b) Consider the pin-supported mild-steel column (E = 200,000 MPa, σ_y = 250 MPa) shown in Figure 4 which is made of a rectangular square hollow section of wall thickness t = 6 mm. The column is restrained in different manners about each axis of bending.

(i) Calculate the load carrying capacity P of the column if the length L = 6 m. (**8 marks**)

(ii) Calculate the length L required for the column to reach its yield stress before it buckles. (**8 marks**)



(a) Major axis bending restraint

(b) Minor axis bending restraint

(c) Section details

Figure 4

- END OF SECTION A -